Simulating the Game of Yahtzee using Python

Group 95 | Katherine Hansen

Georgia Institute of Technology

ISYE 6644: Simulation

Abstract

Yahtzee is a strategic dice game played over 13 rounds with the goal of achieving a high score by gaining points based on various dice combinations. This simulation of the game was developed using Python and prioritized lower section combinations to score points while falling back to upper section categories if necessary. Two main strategies were created which focused on either rolling multiple dice of the same face value or rolling for the straights categories. The performance of the program was analyzed through a Monte Carlo simulation of 10,000 games. The results showed an average game score of 197.14 with a 95% confidence interval of [196.22, 198.06]. Also shown was that the upper section bonus being rarely awarded, in 3.53% of games, and a Yahtzee being scored in 24.38% of games. The middling scores reflected the conservative strategy used and may be improved by taking a riskier approach.

Background

Instructions

Yahtzee is a game using five dice to roll for various combinations to score points over the course of 13 rounds. Each round, all five dice are rolled and may then be re-rolled two more times. After the initial roll, any or all dice may be kept aside with only the remaining dice re-rolled; the process is repeated for the final roll. The ending dice combination is then scored on a scorecard shown in Appendix A. The Yahtzee scorecard has 13 categories split between an upper section and a lower section. After each round, a score must be entered for one of the categories after which it cannot be used again for the remainder of the game. If a dice combination does not satisfy any available scoring categories, a zero must be entered into an unscored category (Milton Bradley Company, 1996).

The upper section of the scorecard consists of the categories Aces, Twos, Threes, Fours, Fives, and Sixes. To score in one of these categories, sum only the dice that match the category's number and enter the total on the scorecard. An additional 35-point bonus is awarded at the end of the game if the total score of the upper section is at least 63 points (Milton Bradley Company, 1996).

The lower section of the scorecard has seven categories. 3 of a Kind is scored if there are at least three dice of the same number as the total of all five dice; 4 of a Kind is scored similarly for when there are at least four dice of the same number. Full House is scored if the dice have three of the same number and two of another different number as 25 points. Small Straight is scored if the dice have a sequence of four numbers as 30 points; Large Straight is scored similarly for a sequence of five numbers as 40 points. A Yahtzee is worth 50 points and is when all five dice show the same number. Chance is scored as the total of any five dice regardless of the combination (Milton Bradley Company, 1996).

If a Yahtzee is rolled after a Yahtzee category was previously scored, a Yahtzee Bonus of 100 points is rewarded but if the Yahtzee category was previously scored as a zero, the bonus is not rewarded. In either case, joker rules are applied: the appropriate upper section category must be scored if available; if not available, score in any lower section category where 3 of a Kind, 4 of a Kind, and Chance as the sum of all dice, while Full House, Small Straight, and Large Straight score as their standard point value. If the appropriate upper section is unavailable and the lower section is filled, the first available upper section category must be scored as zero (Milton Bradley Company, 1996).

Approach

This Yahtzee simulation was implemented using Python and involves simulating dice rolling, scorekeeping, and strategy logic. Following a description of the code development, the performance is analyzed using Monte Carlo simulation discussion. Then strategies and ways to improve game scores are discussed.

Methods

Initialize Game

The simulation game starts with an empty list and dictionary to track the dice hand. A dictionary is also created for the scorecard with the keys being the 13 categories and the values as a list of three values; the first value is a Boolean initialized as false to mark whether the category has been scored, then the minimum score the category is worth if not forfeited, and lastly the points earned for the category. Finally, a variable to count bonus Yahtzees is initialized as zero.

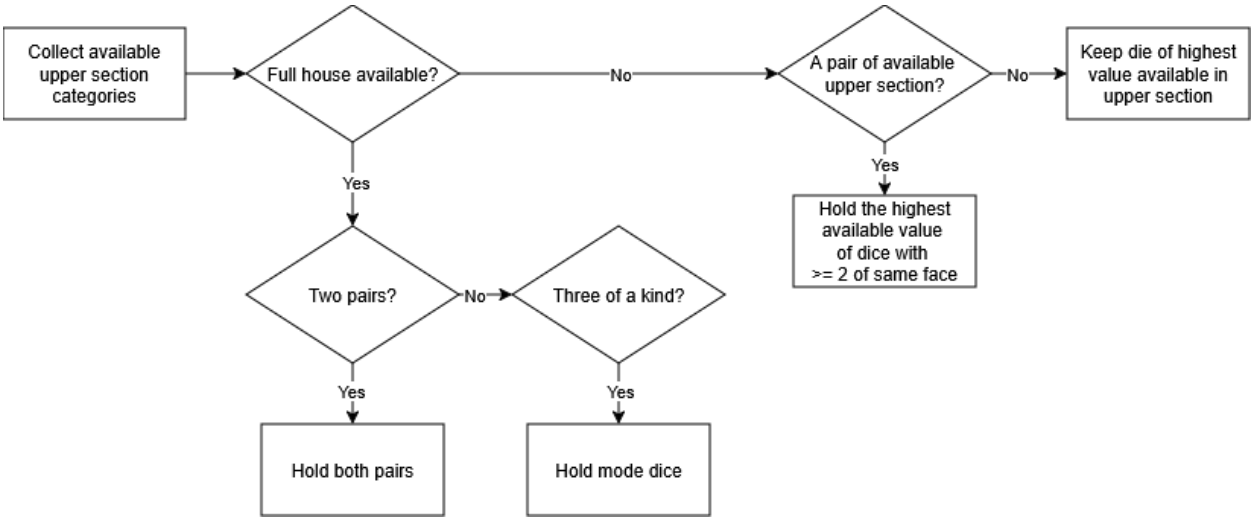
"Of a Kind" Strategy

The function `reroll_threeormore` is used when rolling for all six upper categories 3 of a Kind, 4 of a Kind, and Full House which all aim for dice of the same face value. It takes the current hand and face values and first determines which upper section categories are available and adds the corresponding numbers to a list. Then it checks if there are any numbers with three or more of a kind already. Lastly, the function adds any value that has dice with at least one pair of like numbers to another list.

The function prioritizes the full house, first checking that the category is available and if it is, then checks if there are two distinct pairs of dice in the hand. If both conditions are met, the

function holds the relevant dice and rerolls for a full house. If there are not two pairs of dice but still three of a single face value that is available in the upper section, the function holds the three (or more) of a kind, rerolling the rest. If not rolling for a full house, the function then checks for a three of a kind of an available upper section category number and if not found, it moves on to checking for a pair of dice with an available upper section category number. If multiple pairs are available, it takes the larger value. If no three of a kind or pair values are available in the upper section, all dice get re-rolled. The function then returns the new hand.

Figure 1. Function `reroll_threeormore`.



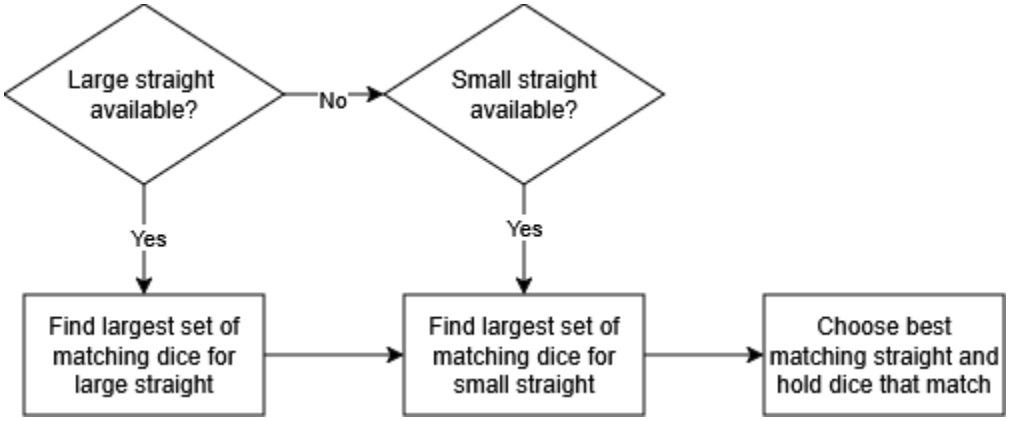
The `reroll_threeormore` function is used when trying to roll multiple dice of the same face. It first checks for a strong full house possibility, then moves onto checking for multiples of a single face value.

Straight Strategy

When rerolling for a small or large straight, the function `reroll_straight` is used. It takes the current hand and sorts it by the face values of the dice then matches to possible straight combinations. The function first checks if the large straight category is available for scoring; if so, it compares the current hand to the two large straight combination possibilities and stores the

best match. Then, if the small straight category is available for scoring, the current hand is compared to the three small straight combination possibilities for the best match. The function cycles through the straight combinations to determine the best match, replacing the best match if the length of the set between the straight combinations is greater than the current best match. After choosing the best match, the function holds the dice in the hand that are part of the straight combination, without duplicates, and re-rolls the rest. The function then returns the new hand.

Figure 2. Function `reroll_straight`.



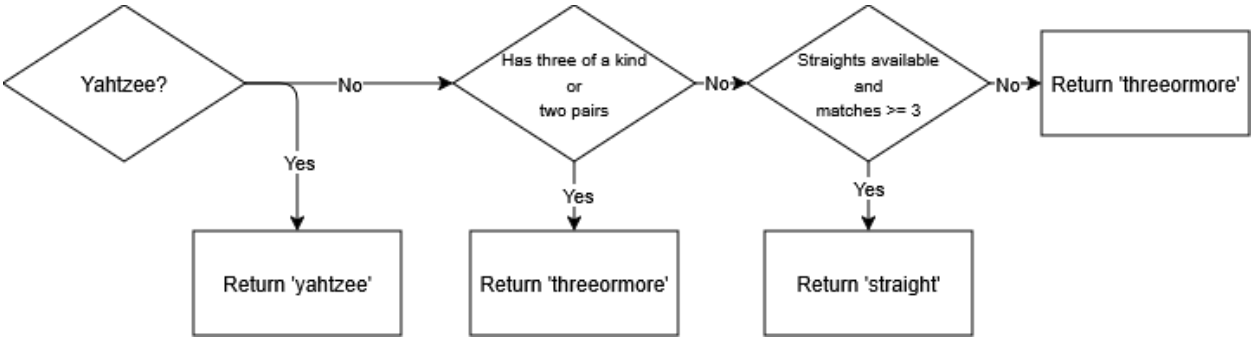
The `reroll_straight` function first checks for large straights and then small straights. It chooses the best match from either straight category based on the number of dice that fit the straight combination.

Strategy Choice

The `choose_strat` function determines which of the previous two strategies to use. It looks at the current hand and first checks for a Yahtzee; if the hand is a Yahtzee, then the function returns ‘yahtzee’. Otherwise, the function prioritizes the three of a kind strategy by checking for multiples of the same face values and if there is a three of a kind or two distinct pairs, the function returns ‘threeormore’. Lastly, it checks for straights. If the hand matches a straight

combination with at least three dice, the function returns ‘straight’. If none of the previous conditions are met, the function defaults to ‘threeormore’.

Figure 3. Function `choose_strat`.

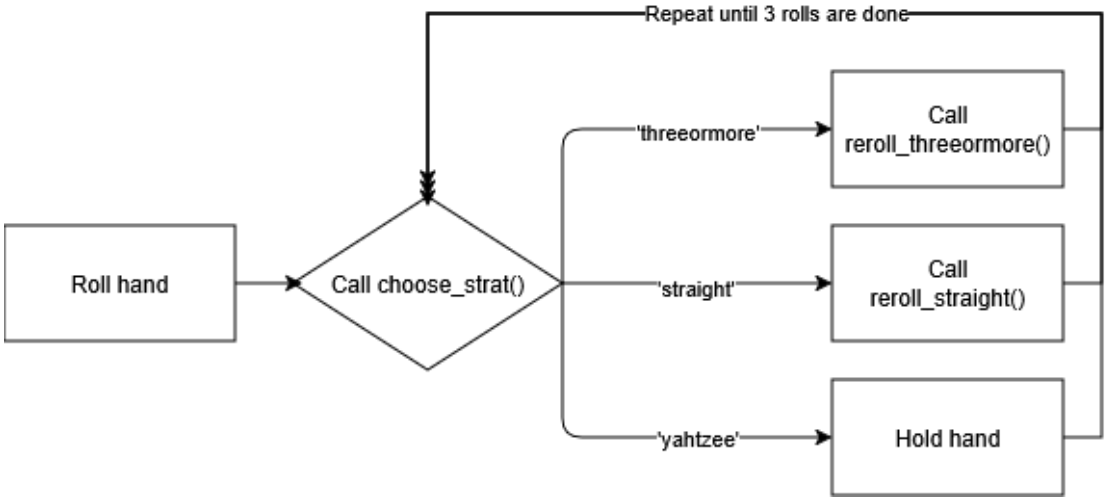


The `choose_strat` function returns which strategy to use for the dice re-roll. It begins by checking for a Yahtzee, then a promising three of a kind or better, then for a straight. If no strong matches are found, the function defaults to the three of a kind strategy.

Play Round

A round is played with the `play_round`, beginning by rolling all five dice. For the next two rolls, the `choose_strat` function is called; if ‘threeormore’ is returned then the `reroll_threeormore` strategy is used, else if ‘straight’ is returned the `reroll_straight` strategy, and if ‘yahtzee’ is returned then no dice are re-rolled.

Figure 4. Function `play_round`.



The `play_round` function goes through all three dice rolls, choosing the appropriate strategy in between each roll.

Scoring

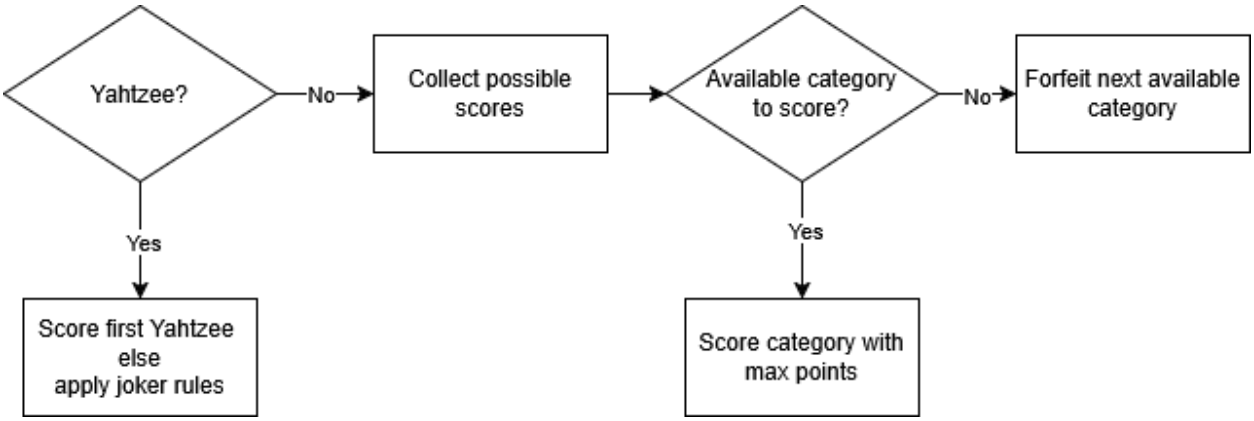
Scoring is handled with a set of three functions: collecting possible scoring options, handling joker rules, and choosing the category to score. The `score_round` function assembles non-zero scoring categories. The function starts by checking for a Yahtzee and invoking the joker rules if applicable. Then the function evaluates the upper section categories and lower section categories. The function adds all the possible scoring category values to a dictionary that gets passed to the `score_best` function.

If the joker rules are invoked, then the `joker_rules` function is called. Following in accordance with the game instructions, first the appropriate upper section category is used if available, otherwise the first available lower section category is used. If all the lower section categories are filled, then the first available upper section category is scored zero.

After the `score_round` and `joker_rules` functions are handled, the `score_best` function uses the previously collected scoring options. If there were no available categories, the

first available category is marked zero. Otherwise, the function chooses the category with the highest score and marks the scorecard appropriately.

Figure 5. Scoring flow.



When scoring a round, the first check is for a Yahtzee and applied joker rules if appropriate. If a Yahtzee is not obtained, the category with the highest point value possible with the current hand is scored. If no category combination is found, then a category must be marked zero.

Play Game

Using the previously defined functions, a full game of Yahtzee can be played using the `simulate_full_game` function. The function begins by playing a round and using the ending hand to determine which category to score. After 13 rounds, the total score is calculated, including additional points for the upper section bonus if applicable, and the bonus Yahtzees if any. This function can be called many times to gather simulation statistics.

Results

The performance of the program was analyzed using 10,000 independent runs of a full 13 round game of Yahtzee. The total score mean across the 10,000 games was 197.14 with a variance of 2,204.95 leading to a 95% confidence interval of [196.22, 198.06]. Additionally, the

number of games that scored the upper section bonus, which rewards 35 additional points if the upper section score exceeds 62 points, was 353 out of 10,000 games (3.53%). Yahtzees were more likely to occur throughout the games with 2,438 out of 10,000 games (24.38%) scoring a Yahtzee. The distribution of the total scores is right skewed with few games reaching scores around and above 300 which can be seen in Appendix B.

Conclusion

Through the development of the Yahtzee game simulation, it became clear the sheer volume of decisions to make: multiple opportunities to choose which dice to hold each round, which category to aim for, and which category to score at the end of every round, for example. The randomness of the dice rolls makes for an exciting new experience every time the game is played.

This implementation was quite conservative in its play style. Before pursuing a strategy, there were often checks in place, particularly when rolling to obtain multiple dice of the same face value, to ensure that the face values were still available in the upper section categories as a safe backup and to avoid forfeiting categories. As the upper section was not actively pursued for high point values, the upper section bonus points were rarely awarded, limiting the total score achieved. Determining which category to score at the end of a round lacked depth by only focusing on obtaining the highest score for that current round. A better strategy might include taking a lower score in earlier rounds to give more chances to score higher rounds later or take larger risks at the beginning of the game and becoming more conservative towards the end.

References

Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. Computing in Science & Engineering, 9(3), 90–95.

Milton Bradley Company. (1996). *Yahtzee: Instructions – 1 or More Players*.

Appendix A



Yahtzee®

NAME _____

UPPER SECTION	HOW TO SCORE	GAME #1	GAME #2	GAME #3	GAME #4	GAME #5	GAME #6
Aces = 1	Count and Add Only Aces						
Twos = 2	Count and Add Only Twos						
Threes = 3	Count and Add Only Threes						
Fours = 4	Count and Add Only Fours						
Fives = 5	Count and Add Only Fives						
Sixes = 6	Count and Add Only Sixes						
TOTAL SCORE	→						
BONUS If total score is 63 or over	SCORE 35						
TOTAL Of Upper Section	→						
LOWER SECTION							
3 of a kind	Add Total Of All Dice						
4 of a kind	Add Total Of All Dice						
Full House	SCORE 25						
Sm. Straight Sequence of 4	SCORE 30						
Lg. Straight Sequence of 5	SCORE 40						
YAHTZEE 5 of a kind	SCORE 50						
Chance	Score Total Of All 5 Dice						
YAHTZEE BONUS	✓ FOR EACH BONUS						
	SCORE 100 PER ✓						
TOTAL Of Lower Section	→						
TOTAL Of Upper Section	→						
GRAND TOTAL	→						

©1982, 1990, 1996 Milton Bradley Company. All Rights Reserved. E6100

Appendix A. A full scorecard to play six games.

Appendix B



Appendix B. The distribution of the total scores output from the 10,000 simulations. The chart was created using the Python package Matplotlib.